The Fusion Reaction

Fusion Reactions:

With the discovery of fission as a sustainable means for energy production, the obvious next question was: Can we do this to lighter elements?

The answer is No. Let's see why:

Clearly, we need to supply energy for this reaction.

$${}_0n^1 + {}_1H^2 \longrightarrow {}_1H^1 + 2({}_0n^1) + E \quad ; \; E = -2.23 \text{ MeV}$$

However, if we try fusion:

$${}_0n^1 + {}_1H^2 \longrightarrow {}_2He^3 + e^- + E \quad ; \; E = +6.27 \text{ MeV}$$

This reaction with positive energy is feasible. However, the reaction needs a neutron to start and the end products do not give us a neutron. Hence, due to the lack of self-sustainability, this reaction is not feasible.

These energies are calculated using $E = mc^2$.

Q: The full relativistic form of energy is: $E^2 = m^2 + p^2$, why do we not account for the momentum part, is it because the momentum is conserved?

The self-sustainability problem can be fixed by swapping the neutron out for a light-element and then providing a steady supply of that element.

This creates another problem: when the nuclei get that close to react, the electrostatic forces between the two nuclei are strong and repulsive in nature. This is what causes difficulty in nuclear fusion. We will neglect this problem.

Q: Is there no other way to incite fusion?

The main characters in our story of fusion reactions are:

${}_0n^1 \rightarrow n$ ${}_1H^1 \rightarrow p$ ${}_1H^2 \rightarrow D$ ${}_1H^3 \rightarrow T$ ${}_2H^3 \rightarrow He^3$

How the reactions:

① The D-D reaction:

This is the most desireable reaction due to deuterium being virtually infinite.

However, it is also the toughest reaction to initiate

However, it is also the hardest reaction to initiate.

$D + D \rightarrow He^3 + n + 3.27\ MeV$ $\quad 78 \times 10^6\ MJ/kg\ (D)$

$D + D \rightarrow T + p + 4.03\ MeV$ $\quad 96 \times 10^6\ MJ/kg\ (D)$

② The D - He^3 reaction:

The advantage of this reaction is that it deals with only charged particles at the end which is much more efficient for energy extraction.

The energy released is also very impressive.

$D + He^3 \rightarrow \alpha + p + 18.3\ MeV$ $\quad 351 \times 10^6\ MJ/kg$

The problem with this reaction is that it is hard to initiate (less so than D-D) and He^3 is not readily available.

③ The D-T reaction:

It is the easiest reaction to initiate and creates a great amount of energy.

$D + T \rightarrow \alpha + n + 17.6\ MeV$ $\quad 338 \times 10^6\ MJ/kg$

The problems associated with this reaction are:

- No natural Tritium.
- Radioactivity of neutrons.
- Radioactivity of Tritium (Half life: 12.6 yrs)

The radioactivity of neutrons and Tritium is already a problem that has been solved in the study of nuclear fission.

The solution to the low supply of Tritium is the Tritium breeder:

$_3Li^6 + n\ (slow) \rightarrow \alpha + T + 4.8\ MeV$

$_3Li^7 + n\ (fast) \rightarrow T + \alpha + n - 2.5\ MeV$

The $_3Li^6$ reaction is easier to initiate and better for us (+ve energy), thus we need a way to slow down the neutrons.

We also need a neutron multiplier if we want the reactor to sustain itself since a few neutron losses are inevitable.

The D-T reaction will be granted most of our focus from here on.

Energy Partition in Fusion Reactions

Most of the released energy is in the form of kinetic energy (The rest: γ-rays)

It is thus important to know how much energy does each end product carry.

Assuming the initial fusing particles to be at rest (w.r.t End product velocities)

we get the 2 conservation equations:

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = E \quad ; \quad m_1 v_1 + m_2 v_2 = 0$$

which gives us nicely solvable equations:

$$\frac{1}{2} m_1 v_1^2 = \frac{m_2}{m_1 + m_2} E \quad ; \quad \frac{1}{2} m_2 v_2^2 = \frac{m_1}{m_1 + m_2} E$$

The ligher particle will carry most of the energy.

For the D-T reaction we get:

$$D + T \longrightarrow \alpha\,(3.5\,\text{MeV}) + n\,(14.1\,\text{MeV})$$

Q: what if there are more than 2 components?